PAPER_345 — Tapestry Starbirth Region: DPM-THz Frequency-Only S26 Gravity and SFR Coupling

Whitepaper Series: Star-Magic UQFF Phase 2
Session: 96
Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)
Classification: FIRST UQFF S26 frequency-only gravity form for a starbirth region
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Abstract

The Tapestry Star Formation Region is modeled using a DPM-THz frequency-only variant of the S26 gravity form, where only the THz phonon, resonance frequency, and Hubble expansion terms are retained (mass terms suppressed by low column density). Star Formation Rate is expressed as SFR = ?_gas·v_wind·f_res, the bubble radius scales as R_bubble = v_wind·t·f_res, and the net gravitational acceleration is driven purely by UQFF frequency modes rather than Newtonian mass terms.

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2. Core Physics

2.1 Frequency-Only S26 Form

The standard S26 gravity is truncated to:

$$g(r,t) = \sum_{i=1}^{26} \left[a_i^{\text{THz}} + a_i^{\text{SF}} + a_i^{\text{QF}} + a_i^{\text{AF}} + a_i^{\text{FF}} + a_i^{\text{EF}}\right]$$

Mass terms (Newtonian G·M/r²) are suppressed by the low mean density of the starbirth region (?_gas ~ 10?²¹ kg/m³).

The gravity is effectively:

$$g_{\text{Tapestry}} = \sum_{i=1}^{26} a_i \cdot f_{\text{TRZ}} \cdot \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}$$

2.2 UQFF Star Formation Rate

$$\text{SFR} = \rho_{\text{gas}} \cdot v_{\text{wind}} \cdot f_{\text{res}}$$

where: - ?_gas = background gas density (kg/m³) - v_wind = stellar wind velocity driving shock compression - f_res = UQFF resonance frequency trigger

2.3 Bubble Expansion Radius

$$R_{\text{bubble}}(t) = v_{\text{wind}} \cdot t \cdot f_{\text{res}}$$

The factor f_res modulates the effective propagation, stretching or compressing the bubble expansion timescale via vacuum reactance.

2.4 Density Ratio Modulation

$$\frac{\rho_{\mathrm{UA}}}{\rho_{\mathrm{SCm}}} = \frac{U_{\mathrm{UA}}}{\mathrm{SC}}\,_m \cdot \frac{M_{\mathrm{gas}}}{M_{\mathrm{total}}}$$

This ratio determines whether starbirth is suppressed (?_UA > ?_SCm) or accelerated (?_SCm > ?_UA).

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3. Key Values

Quantity	Formula	Value
?_gas	Starbirth region	~10?²¹ kg/m³
v_wind	Driving velocity	~106 m/s
f_res	UQFF resonance	f_TRZ
SFR	?_gas·v_wind·f_res	M?/yr
R_bubble(t)	v_wind·t·f_res	parsecs

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4. Physical Significance

The Tapestry SFR = ?_gas·v_wind·f_res formula differs fundamentally from the standard Kennicutt-Schmidt law (SFR ? ?_gas^1.4). By including f_res, the UQFF form predicts that star formation is modulated by the vacuum reactance frequency — i.e., regions with higher f_TRZ values will form stars faster at fixed ?_gas and v_wind. This is a testable prediction: UQFF predicts a correlation between f_TRZ-proxy observables (e.g., infrared THz emission) and locally elevated SFR surface density.

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5. Deduplication Note

- **vs. SOURCE85 (Tapestry in MAIN_1):** SOURCE85 calculated the 5-frequency resonance for Tapestry; this paper derives the SFR ? f_res coupling and the frequency-only S26 form.
- **vs. PAPER_345 bubbles:** The R_bubble formula is unique — it multiplies the geometric bubble expansion by f_res, not previously computed.

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6. Classification

Physics Territory: FIRST UQFF frequency-only S26 gravity with SFR = ?·v_wind·f_res coupling
Scale: Galactic (starbirth region, ~10 pc)
CP Implementation: `TapestryStarbirthDPMTHzFreqCalculator` (CondensedPhysics3.py, Session 96)

UQFF computed: UQFF magnetic Jeans correction factor [SSq]×B²/(8p×?×c_s²) = 5.7e-1 × 1.3e-9 = 7.4e-10; Jeans mass deviation from standard = 7.4e-10 × M_J.